

LegalEase: AI-Powered Legal Document Generator

Project Documentation Phase

1. Team Details

Team Name: LegalEase: AI-Powered Legal Document generator

Team Members:

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2. Project Overview

LegalEase is an AI-powered legal document generator designed to simplify the preparation of legal document drafts. The system accepts basic information from the user and uses Generative AI to create a structured document that can be previewed, edited, and downloaded.

3. Problem Statement

Preparing legal documents manually can be time-consuming and may require knowledge of legal formats. Users often need to organize information such as parties, terms, and dates into a proper document structure. LegalEase aims to reduce this effort through an AI-assisted document generation workflow.

4. Objectives

- Develop an AI-powered legal document generation application.
- Provide a simple and user-friendly interface.
- Generate drafts based on user-provided information.
- Allow users to preview and edit generated content.
- Provide PDF and Word download options.

5. Technologies Used

- Python
- Streamlit
- FastAPI
- Uvicorn
- Google Gemini / Google Gen AI SDK
- python-docx
- PDF generation library

- Requests
- python-dotenv
- GitHub

6. System Architecture

The system consists of a Streamlit frontend, FastAPI backend, Gemini AI module, and document export components. The frontend collects user information and sends it to the backend. The backend communicates with the Gemini AI module, receives the generated document, and returns it to the frontend for preview, editing, and download.

7. Project Modules

- Frontend Module – provides the Streamlit user interface.
- Backend Module – handles API requests using FastAPI.
- AI Module – generates legal document drafts using Gemini.
- Document Processing Module – creates PDF and Word files.
- Configuration Module – manages environment variables and API configuration.

8. User Inputs

- Document Type
- Parties
- Terms and Conditions
- Effective Date

9. Working Process

- The user selects the required legal document type.
- The user enters party details, terms, and effective date.
- The frontend sends the information to the FastAPI backend.
- The backend validates the request.
- The Gemini AI module generates the legal document draft.
- The generated document is displayed to the user.
- The user can edit the generated content.
- The final document can be downloaded as PDF or Word.

10. API Endpoint

The main document-generation endpoint is POST /generate. It accepts document_type, parties, terms, and dates as request data and returns the generated document content.

11. Key Features

- AI-based legal document generation
- Multiple document types
- Simple user input interface
- Document preview
- Editable generated content
- PDF export
- Word export
- FastAPI-based backend

12. Testing and Validation

The application is tested for frontend operation, API communication, AI document generation, editing, and PDF/Word downloads. Different document types and sample inputs are used to verify the main workflow.

13. Advantages

- Reduces the effort involved in preparing document drafts.
- Provides a simple interface for users.
- Generates structured content using Generative AI.
- Allows users to edit the generated document.
- Provides downloadable document formats.

14. Limitations

- Generated documents are drafts and may require legal review.
- The application depends on availability of the Gemini AI service.
- Output quality depends on the information provided by the user.
- Internet access is required for AI generation.

15. Future Enhancements

- Add more legal document templates.
- Add user authentication and document history.
- Add advanced document formatting.
- Add additional export formats.
- Add multilingual document generation.
- Improve document validation and customization.

16. Security Note

The Gemini API key should be stored securely using environment variables and should not be uploaded to a public GitHub repository. Users should review generated legal documents with a qualified legal professional before using them for real-world legal purposes.

17. Conclusion

LegalEase demonstrates how Generative AI can be integrated with a web application to simplify legal document drafting. By combining Streamlit, FastAPI, and Gemini, the project provides a workflow for generating, previewing, editing, and exporting legal document drafts.